EXPERIMENT 3: COMPARISON OF SGD WITH MOMENTUM VS ADAM OPTIMIZER

Aim:

To compare SGD with Momentum and Adam optimizers on CIFAR-10 using a neural network.

Cell 1 – Import libraries

```
import torch
import torch.nn as nn
import torch.optim as optim
import torchvision
import torchvision.transforms as transforms
import matplotlib.pyplot as plt
```

Cell 2 – Load CIFAR-10

```
device = torch.device("cuda" if torch.cuda.is_available() else "cpu")
transform = transforms.Compose([
    transforms.ToTensor(),
    transforms.Normalize((0.5,), (0.5,))
])
trainset = torchvision.datasets.CIFAR10(
    root='./data', train=True, transform=transform, download=True
)
trainloader = torch.utils.data.DataLoader(
    trainset, batch_size=128, shuffle=True
)
```

Cell 3 – Define the MLP

```
class MLP(nn.Module):
    def __init__(self):
        super().__init__()
        self.flatten = nn.Flatten()
        self.net = nn.Sequential(
            nn.Linear(3*32*32, 256),
            nn.ReLU(),
            nn.Linear(256, 10)
        )
    def forward(self, x):
        x = self.flatten(x)
        return self.net(x)
```

```
Epoch 1: Loss = 1.8802, Accuracy = 32.63%
Epoch 2: Loss = 1.6861, Accuracy = 40.20%
Epoch 3: Loss = 1.6100, Accuracy = 43.19%
Epoch 1: Loss = 1.8739, Accuracy = 32.78%
Epoch 2: Loss = 1.7069, Accuracy = 39.34%
Epoch 3: Loss = 1.6331, Accuracy = 42.05%
```

Cell 4 – Training function

```
def train(model, optimizer, epochs=10):
    model.to(device)
    loss_fn = nn.CrossEntropyLoss()
```

```
    losses, accuracies = [], []
    for epoch in range(epochs):
        total_loss, correct, total = 0, 0, 0
        model.train()
        for imgs, labels in trainloader:
            imgs, labels = imgs.to(device), labels.to(device)
            outputs = model(imgs)
            loss = loss_fn(outputs, labels)
            optimizer.zero_grad()
            loss.backward()
            optimizer.step()
            total_loss += loss.item()
            _, predicted = outputs.max(1)
            total += labels.size(0)
            correct += (predicted == labels).sum().item()
        avg_loss = total_loss / len(trainloader)
        accuracy = 100.0 * correct / total
        losses.append(avg_loss)
        accuracies.append(accuracy)
        print(f"Epoch {epoch+1}: Loss = {avg_loss:.4f}, Accuracy = {accuracy:.2f}%")
    return losses, accuracies
```

Cell 5 – Train using both optimizers

```
model_sgd = MLP()
sgd = optim.SGD(model_sgd.parameters(), lr=0.01, momentum=0.9)
losses_sgd, acc_sgd = train(model_sgd, sgd)

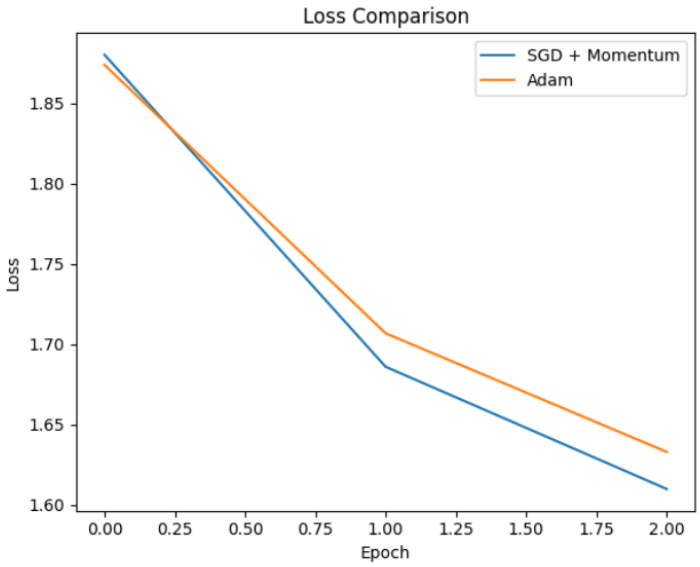
model_adam = MLP()
adam = optim.Adam(model_adam.parameters(), lr=0.001)
losses_adam, acc_adam = train(model_adam, adam)
```

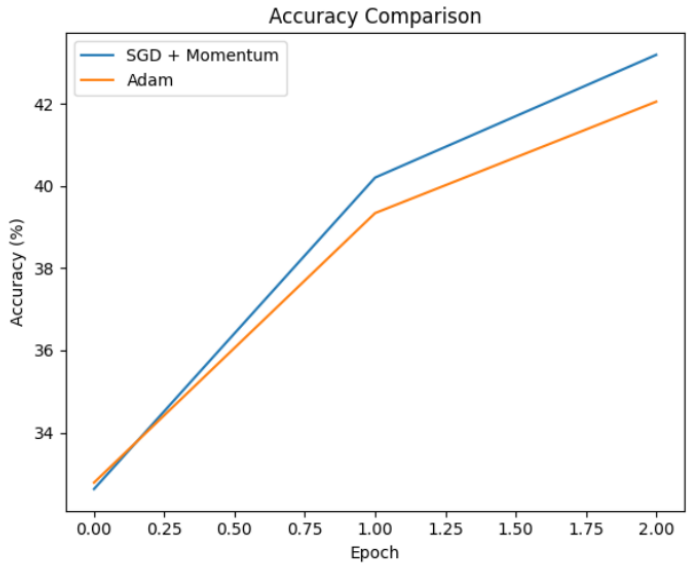
Result: Separate models are trained using SGD with Momentum and Adam.

Cell 6 – Plot comparison

```
plt.figure(figsize=(12, 5))
plt.subplot(1, 2, 1)
plt.plot(losses_sgd, label="SGD + Momentum")
plt.plot(losses_adam, label="Adam")
plt.xlabel("Epoch"); plt.ylabel("Loss")
plt.title("Loss Comparison on CIFAR-10 (MLP)")
plt.legend()

plt.subplot(1, 2, 2)
plt.plot(acc_sgd, label="SGD + Momentum")
plt.plot(acc_adam, label="Adam")
plt.xlabel("Epoch"); plt.ylabel("Accuracy (%)")
plt.title("Accuracy Comparison on CIFAR-10 (MLP)")
plt.legend()
plt.tight_layout()
plt.show()
```





RESULT

SGD with Momentum and Adam were implemented and compared using training loss and accuracy.